

OB.M.S. COLLEGE OF ENGINEERING BENGALURU

Autonomous Institute, Affiliated to VTU



Lab Record

Object Oriented Analysis and Design

Submitted in partial fulfillment for the 5th Semester Laboratory

Bachelor of Technology
in
Computer Science and Engineering

Submitted by:

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Aug-Dec 2025

B.M.S. COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING



CERTIFICATE

This is to certify that the Object-Oriented Analysis and Design(23CS6PCSEO) laboratory has been carried out by G M Kusuma (1BM24CS405) during the 5th Semester Aug-Dec-2025.

Signature of the Faculty Incharge:

NAME OF THE FACULTY: RoopaShree CS

Department of Computer Science and Engineering
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1. Hotel Management System

SRS DOCUMENT

Software Requirements Specification on Hotel Management System

1. Introduction

1.1 purpose

This document specifies the software required for the Hotel Management System. The system will automate hotel operations such as room booking, check-in/check-out, billing and reporting to improve operational efficiency and guest experience.

1.2 Document Conventions

- Bold for heading and imp terms
- Bullet Point for list
- use clear, simple language.

1.3 Intended Audience and Reading Suggestion

- Developers and testers for implementation and validation.
- project managers for planning and tracking
- Hotel staff and stakeholders for understanding capabilities.

1.4 product scope

HMS is a web-based application designed to manage hotel reservation, guest check-in/check-out, payment, user roles and reporting.

1.5 References

- IEEE Std.
- PCI DSS guidelines for payment security.
- GDPR for data protection compliance.

2. Overall Description

2.1 product perspective

The HMS is a standalone web application integrated with payment gateways and notification service.

2.2 product features

- user registration and authentication.
- Room search and booking.
- booking notification and cancellation.
- guest check-in and check-out.
- payment processing and invoicing.
- Report generation.

2.3 user classes and characteristics

Admin: full access, manages sys and reports
Receptionist: Manage bookings and guest queries
Guest: searches and books rooms, manages profile.

2.4 Operation Environment

- modern web browsers on desktop and mobile.
- cloud or on-premise server desktop and mobile.

2.5 Design and Implementation Constraints

- secure payment processing.
- data encryption in transit and at rest.
- Role based access control.

2.6 User documentation

- user manuals and online help.
- Admin and staff training guides.

2.7 Assumptions and Dependencies

- Internet connection for payment and notification.
- Third party APIs availability.

3. Specification Requirements

3.1 functional requirements

- Secure user registration and login.
- Role based access control.
- Room availability search by date, type and price.
- Book, modify and cancel reservation.
- check-in and check-out processing.
- Process payments and generate receipts.
- manage user profiles.

- Admin reports on bookings, revenue and occupancy
- Notification via email/SMS

3.2 External Interface Requirements

- user-friendly web UI
- payment gateway integration (Stripe, Paypal)
- Email and SMS notification API's
- Secure HTTPS communication

3.3 System features

- Room management: add/update/delete rooms and availability
- Booking management: full booking lifecycle handling
- user Management: roles and access controls
- payment processing and Billing
- Reporting Dashboard

3.4 Non-functional Requirements

- Support 100+ Concurrent users
- Response time under 2 seconds
- Data security and encryption
- 99.9% system uptime
- Scalable architecture for future expansion

4. Appendices

4.1 Glossary

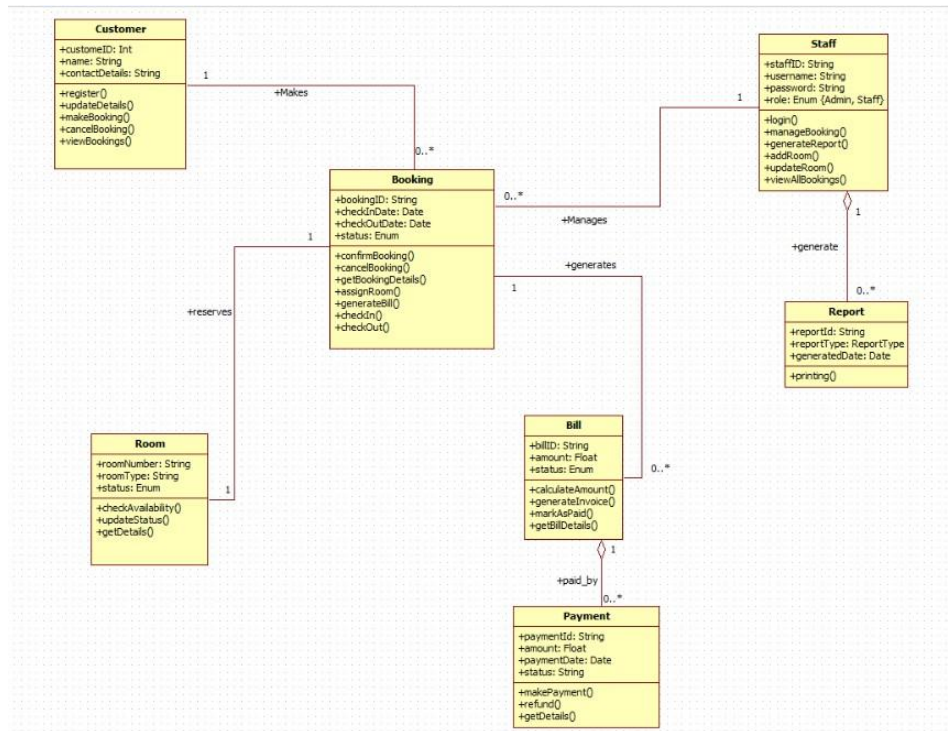
- Booking: Reservations of Rooms
- Check-in: Guest arrival process
- Check-out: Guest departure and payment finalization

4.2 Future enhancements

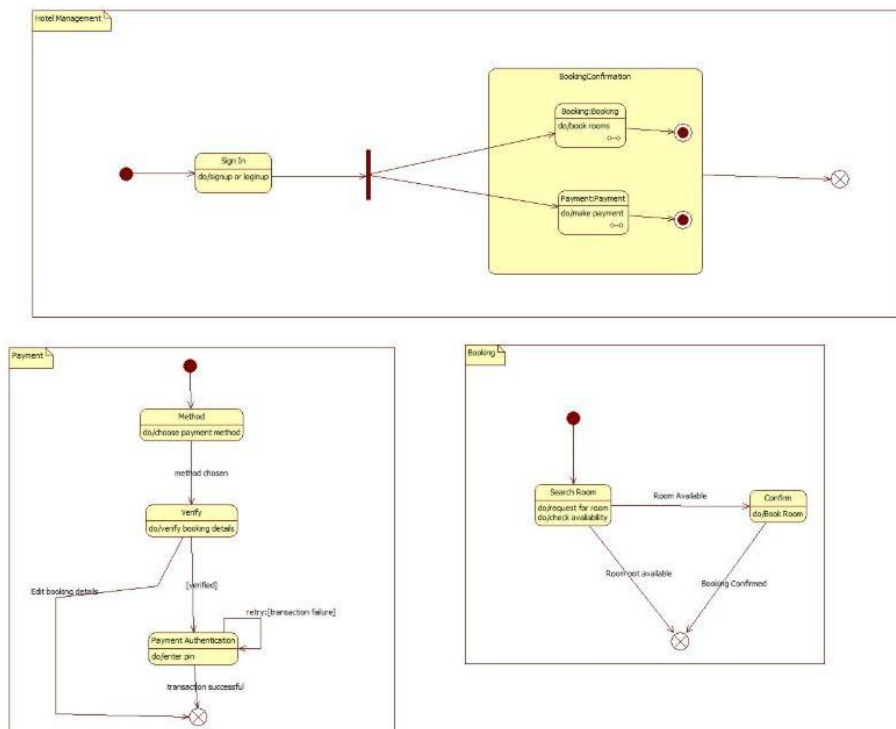
- Mobile apps
- Door lock system
- Loyalty programs
- Multi-language support



CLASS DIAGRAM

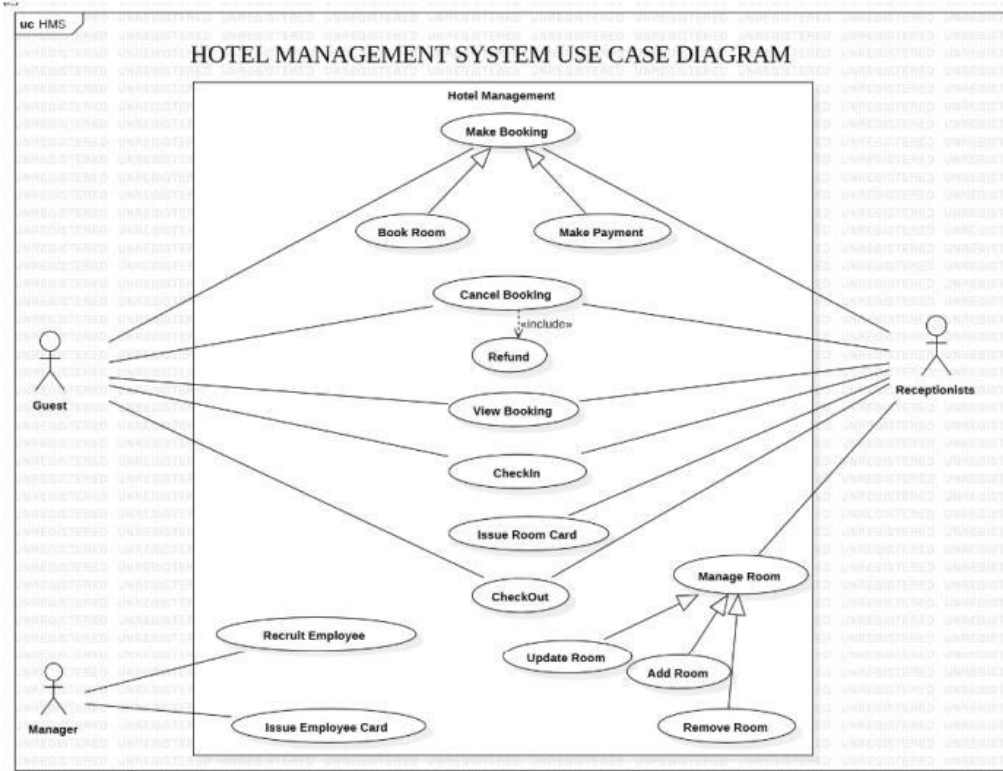


STATE DIAGRAM

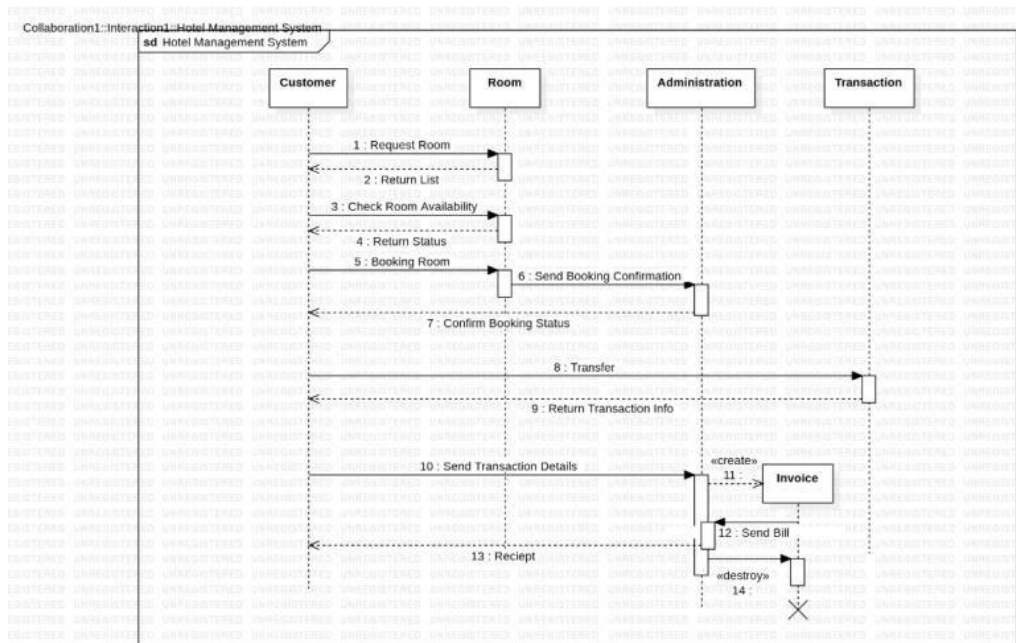


INTERACTION MODELS

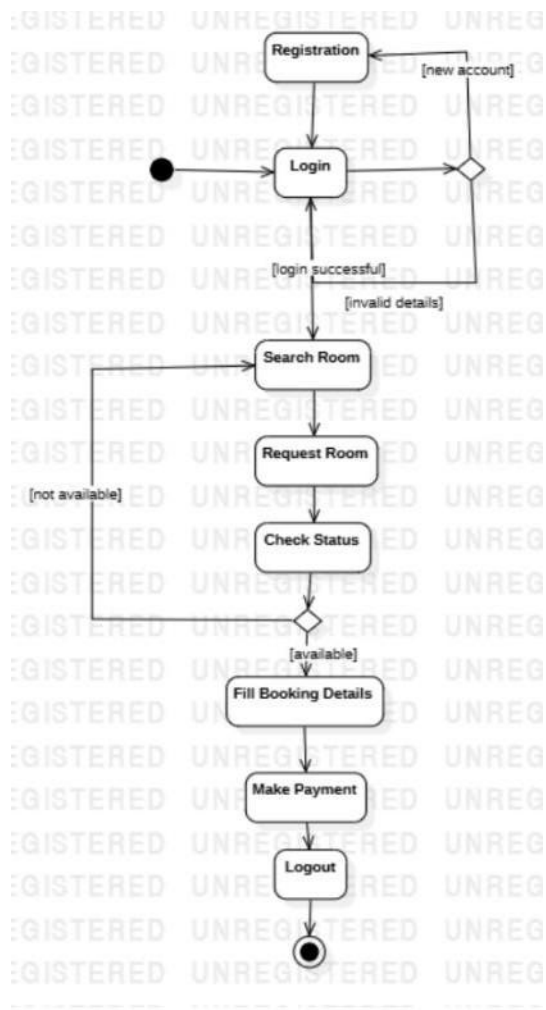
a. Use case model



b. Sequence model



c. Activity model



2. Credit Card Processing

2.1 SRS DOCUMENT

Software Requirements Specification for Credit Card processing system

1. Introduction

1.1 purpose

It is a secure system designed to authorize, process and settle credit card transactions for merchants. This SRS covers the core transaction processing functionalities including authorization, capture, refund and settlement.

1.2 Document Conventions

- Requirements are standalone and do not inherit priority from other requirements.
- IEEE format

1.3 Intended Audience and Reading Suggestions

- This SRS is intended for developers, project managers, testers and stakeholders. Readers are encouraged to start with the specific requirements based on their roles.

1.4 project scope

- It supports authorization, settlements, refunds handling and fraud detection. The system aligns with organizational goals to modernize and secure transaction handling.

1.5 References

Credit card network API guides

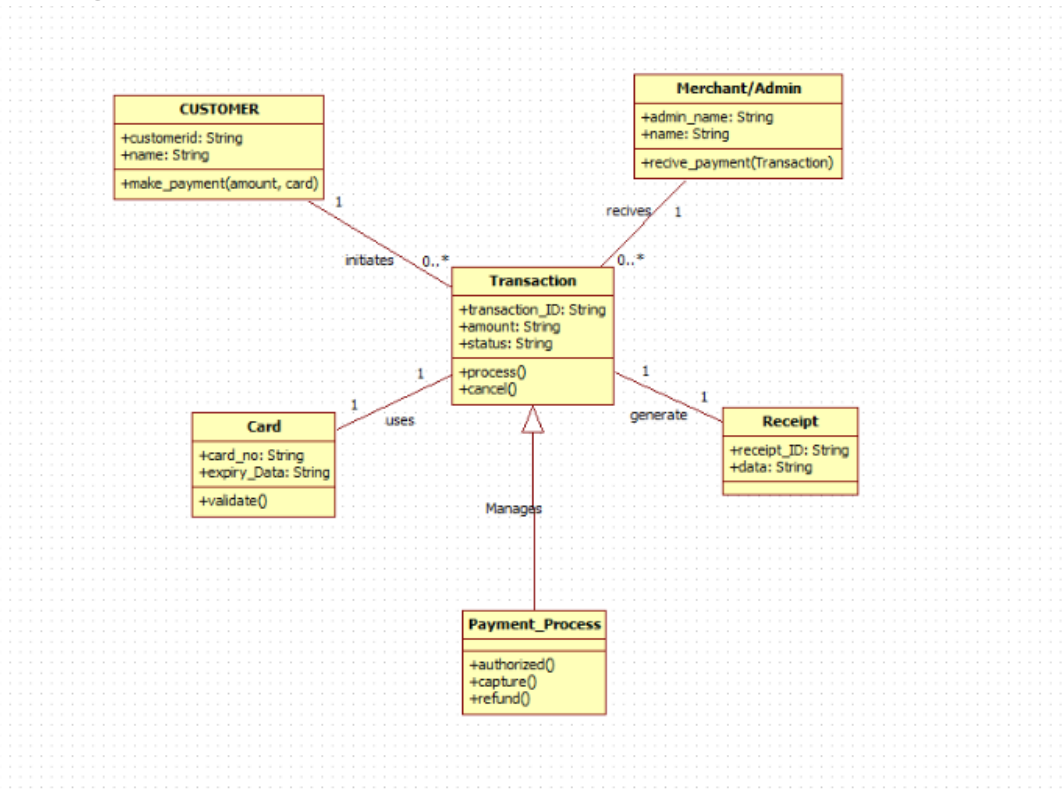
2 Overall Description

2.1 product perspective

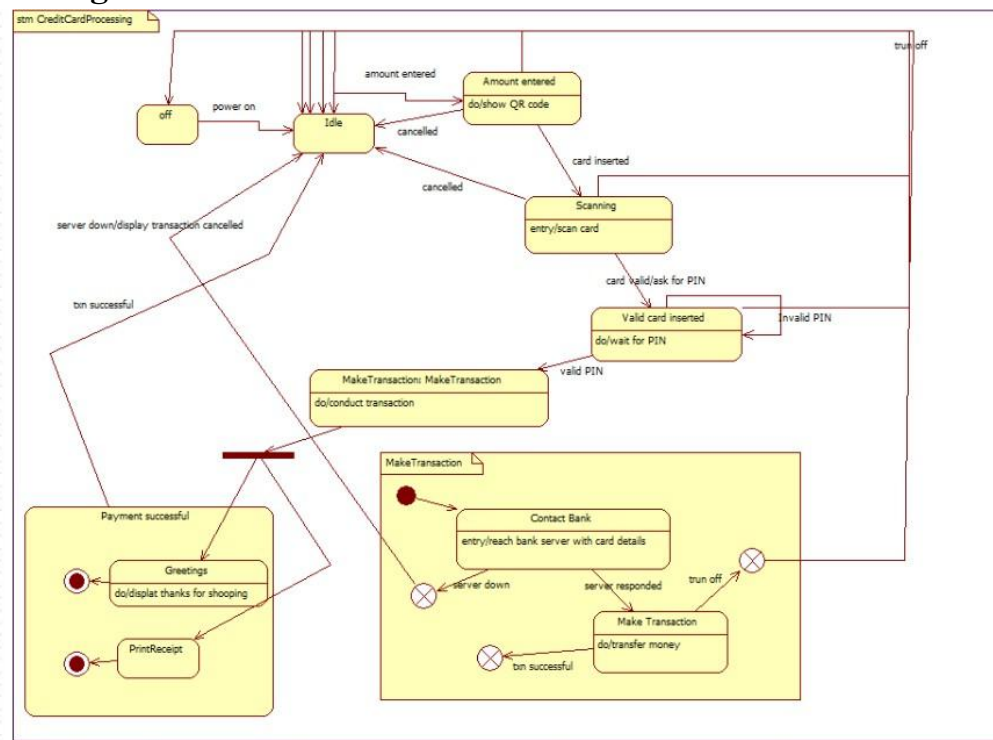
- It is a standalone system replacing older payments system. It interfaces with card networks, banking APIs. The system supports integration with external services through standardized APIs.

- 2.2 product features
 - The system authorizes transactions, captures payments, processes settlements, handles chargebacks and generates reports.
- 2.3 user classes and characteristics
 - user groups include customers, merchants, admin staff and developers.
- 2.4 operating environment
 - The system runs on a cloud platform with databases. 2.0 for secure communication.
- 2.6 user documentation
 - will be provided in both formats.
 - will be integrated in the user interface.
- 2.7 Assumptions and Dependencies
 - The system depends on external card networks, stable APIs, and ongoing access to cloud info. It assumes users access the system through secure channels.
- 3. Specification requirements
 - 3.1 functional requirements
 - authorize transactions within 3 sec
 - capture payment after merchant confirms
 - process merchant-initiated refunds.
 - keep transaction logs for 2 years.
 - 3.2 External Interface Requirement
 - use standard version for payment network messaging.
 - provide RESTful APIs for merchant
 - 3.3 External System features
 - Role based access control
 - Real time monitoring dashboard
 - Automated backup and recovery.
 - 3.4 Non-functional Requirement
 - 99.9% system availability
 - Encrypt all sensitive data.
 - Handles 10,000 transactions/sec

2.3 Class Diagram

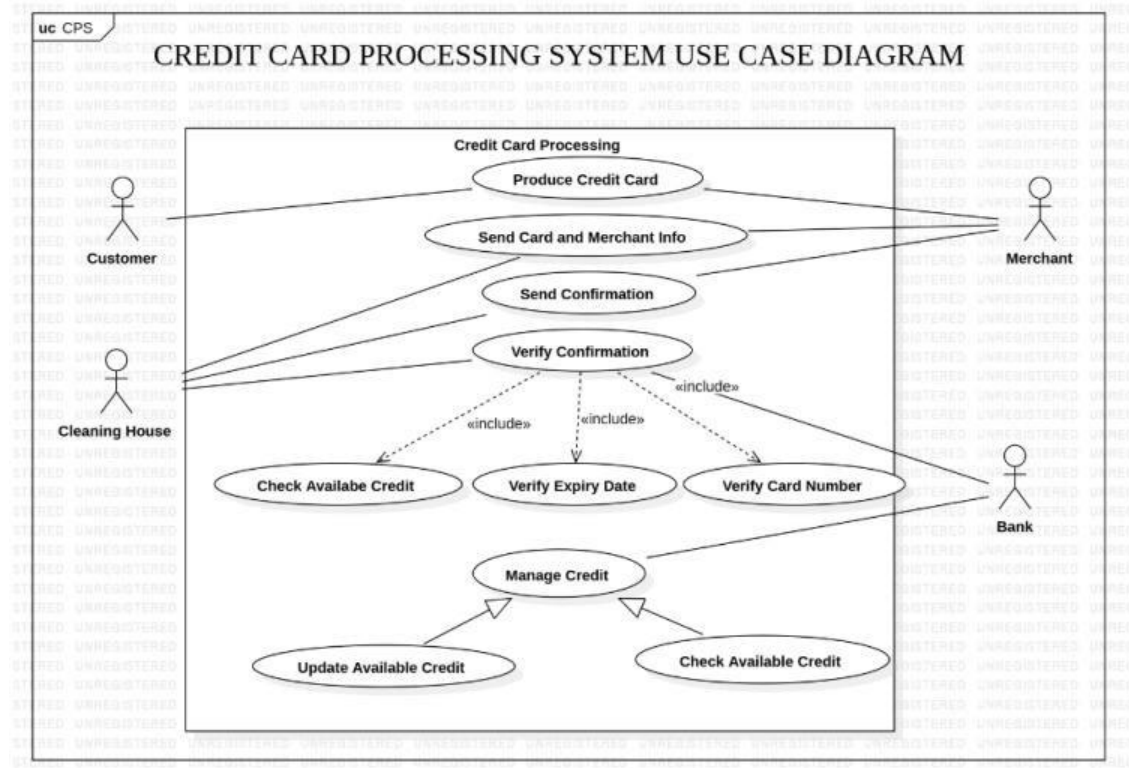


2.3 State Diagram

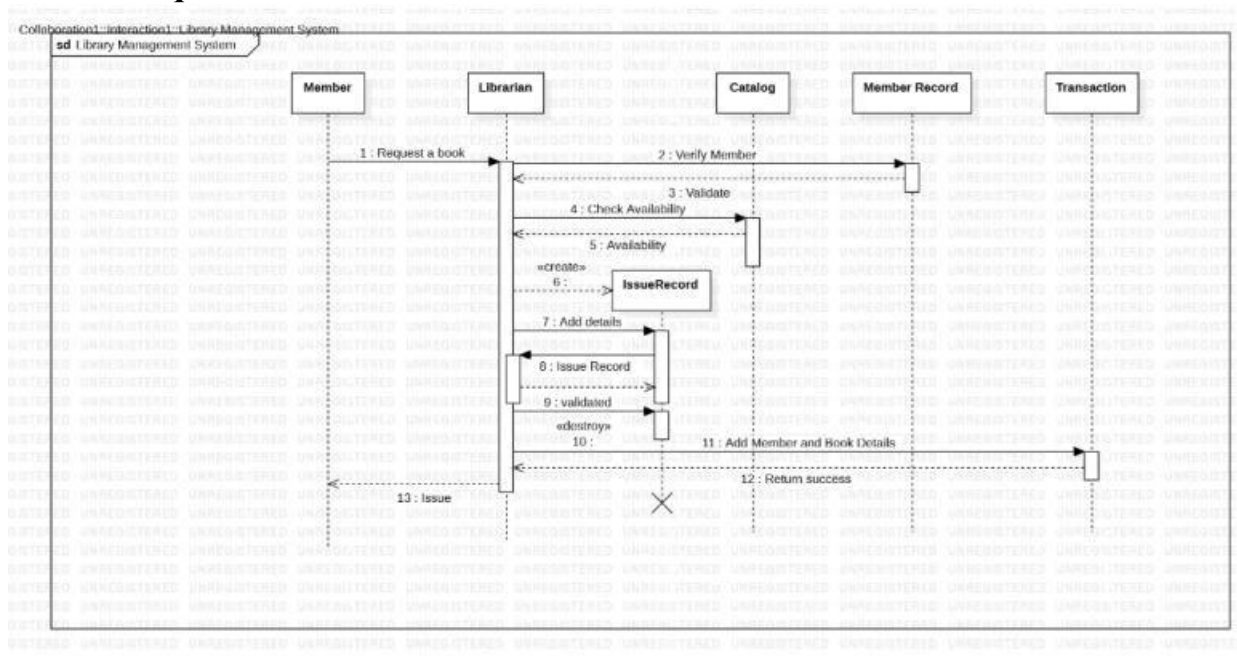


2.4 Interaction Models

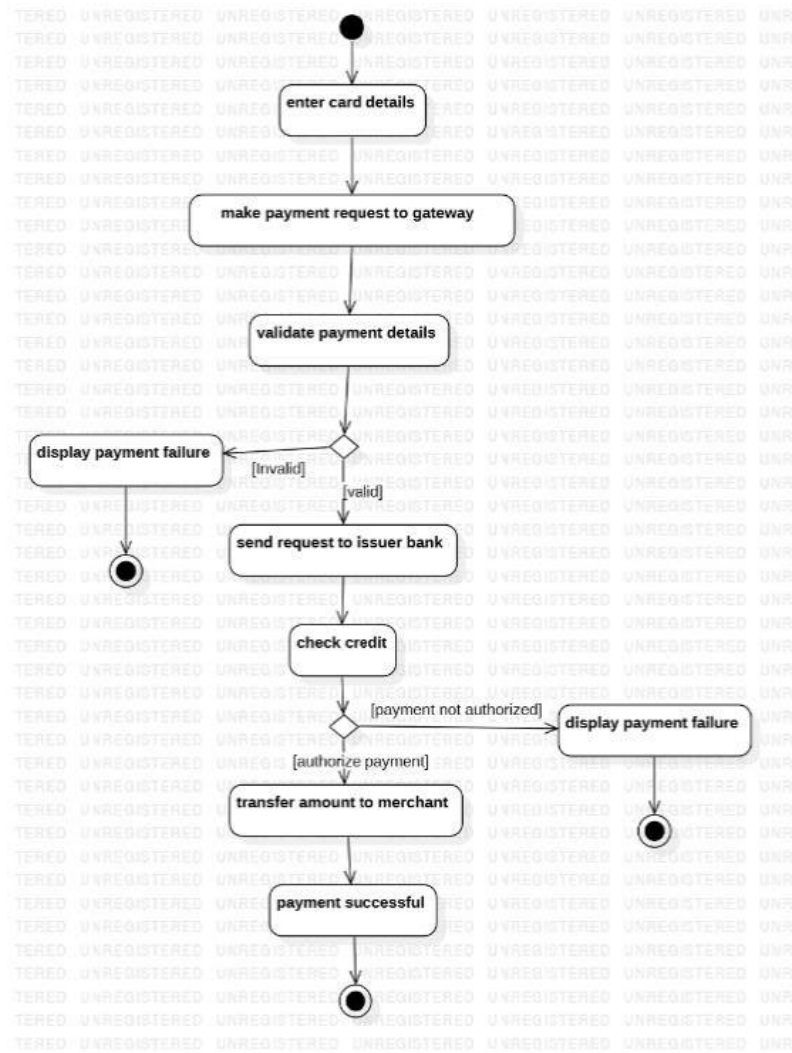
a. Use Case Model



d. Sequence model

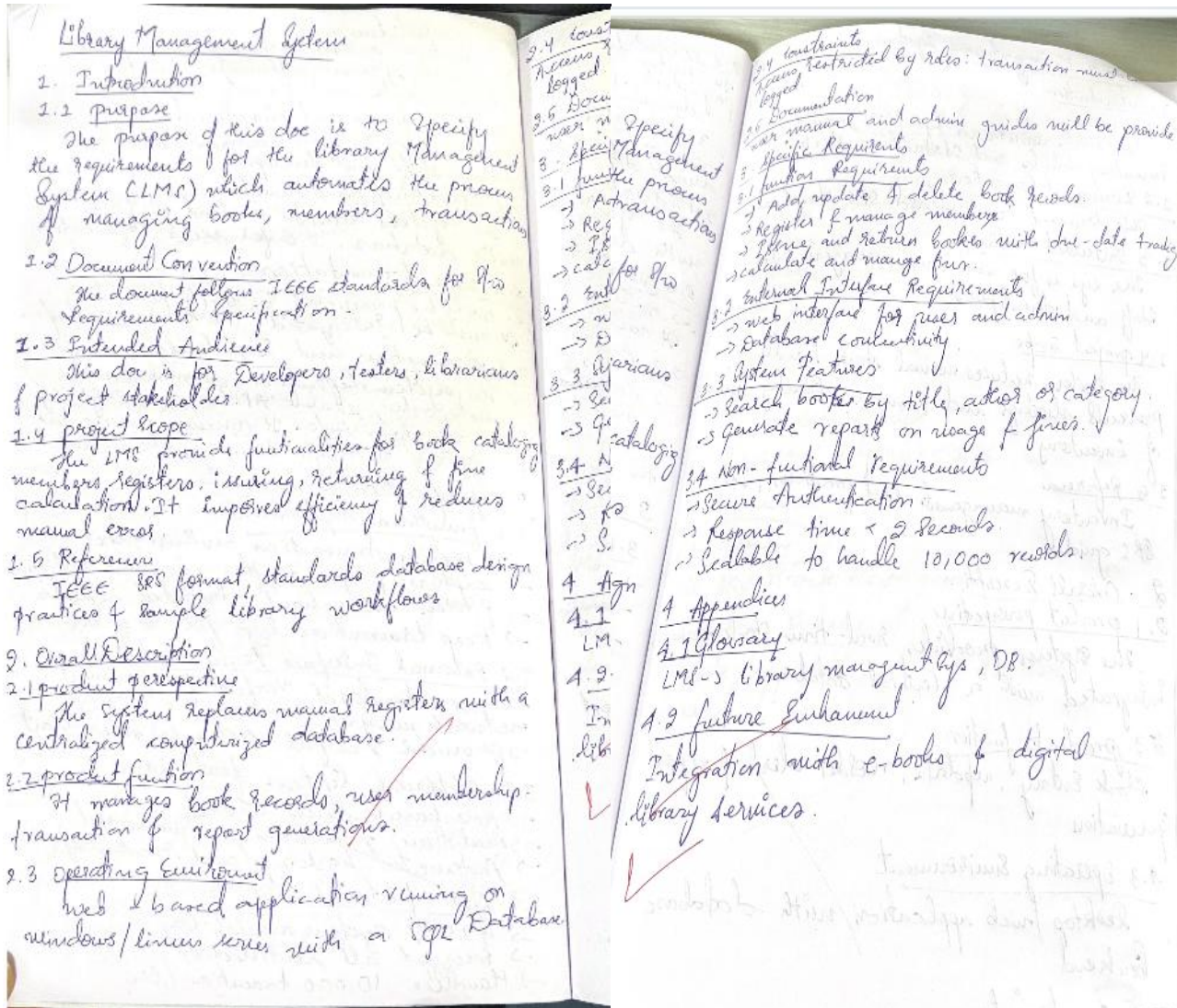


c. Activity Model

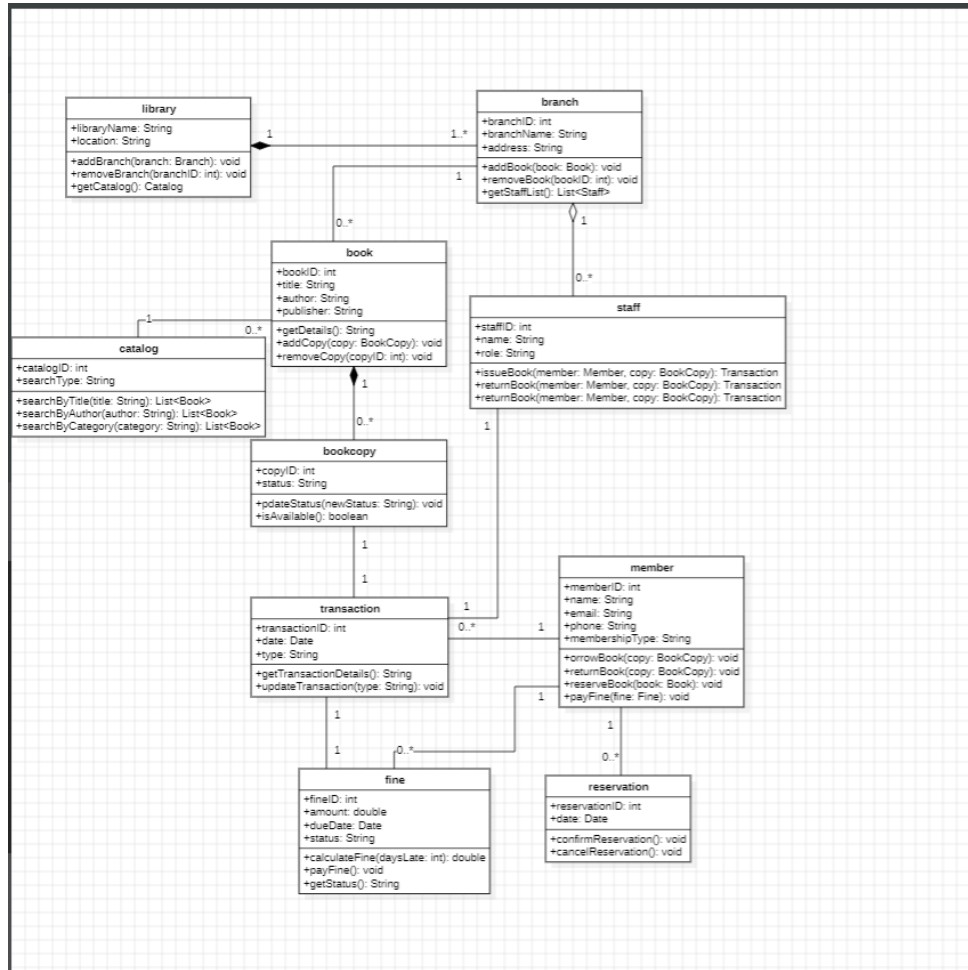


3. Library management system

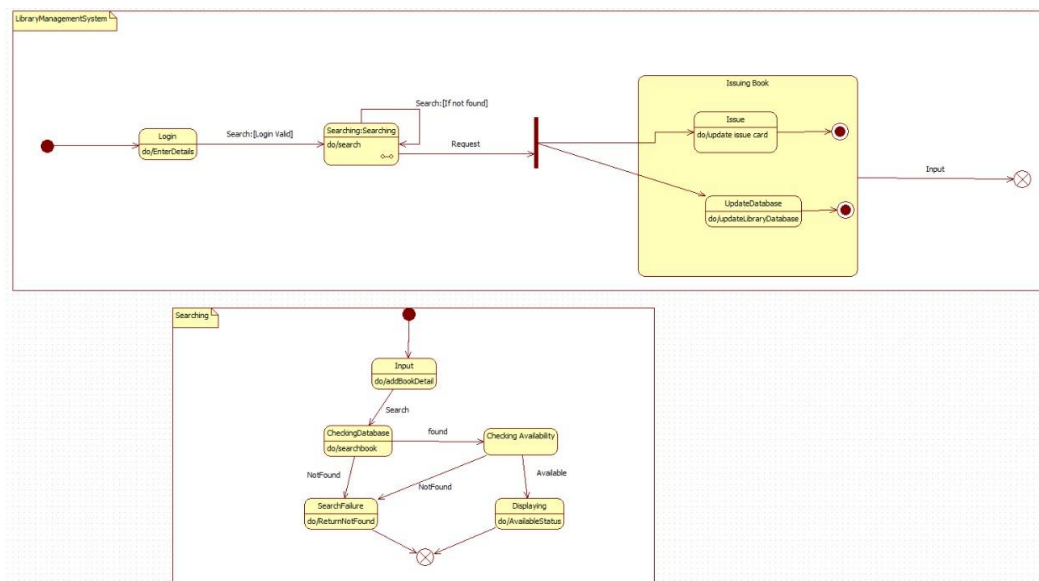
3.1 SRS Document



3.3 Class Model

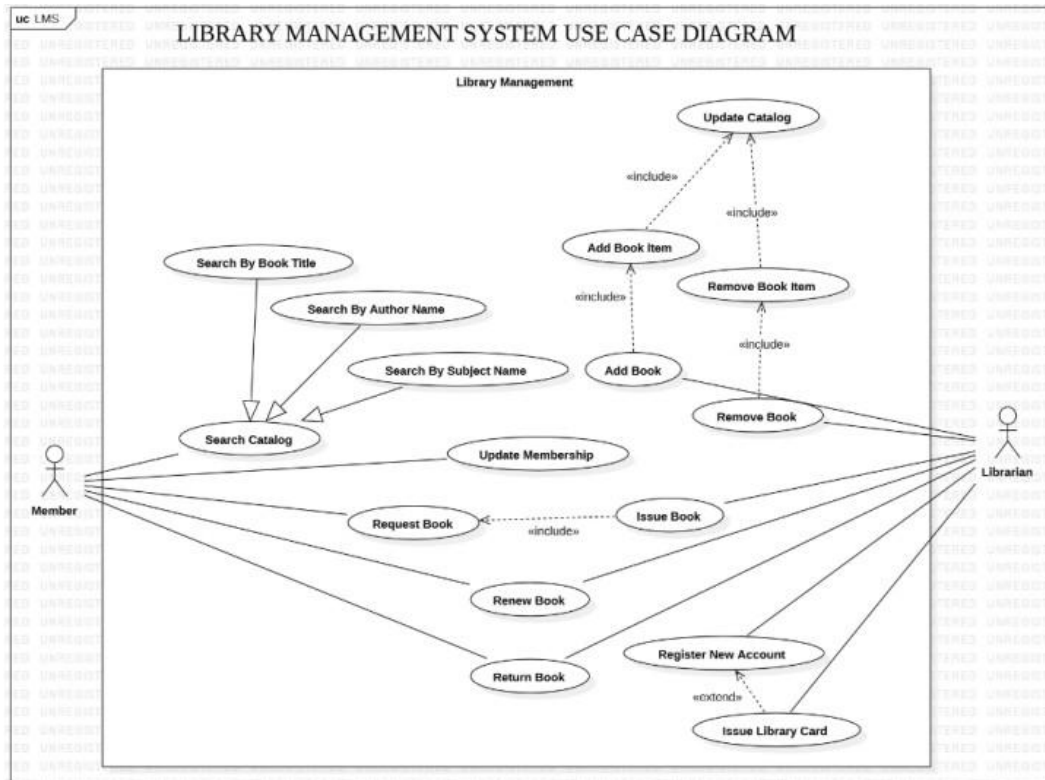


3.4 State model

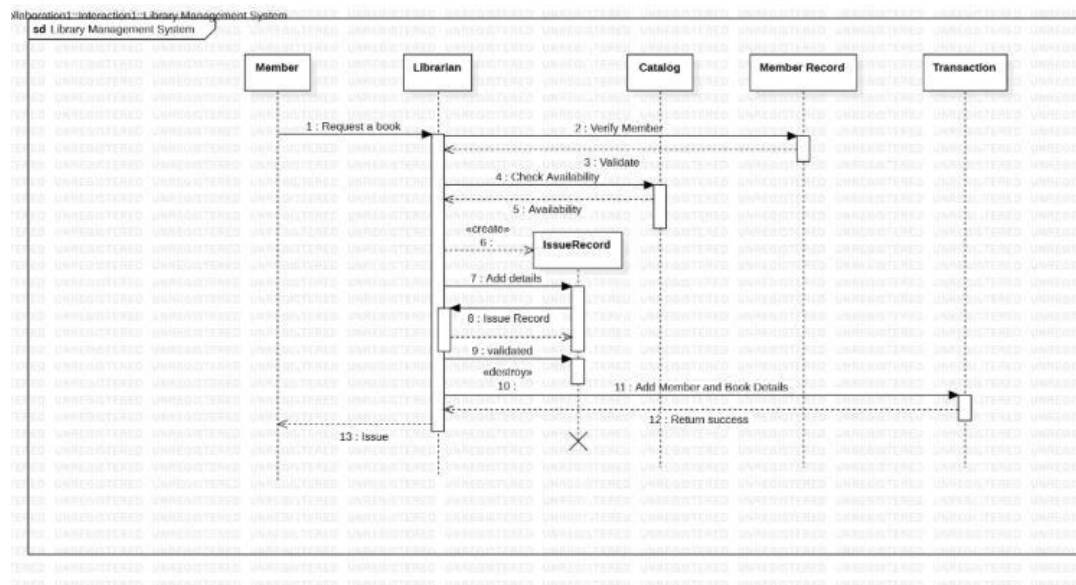


3.5 Interaction Models

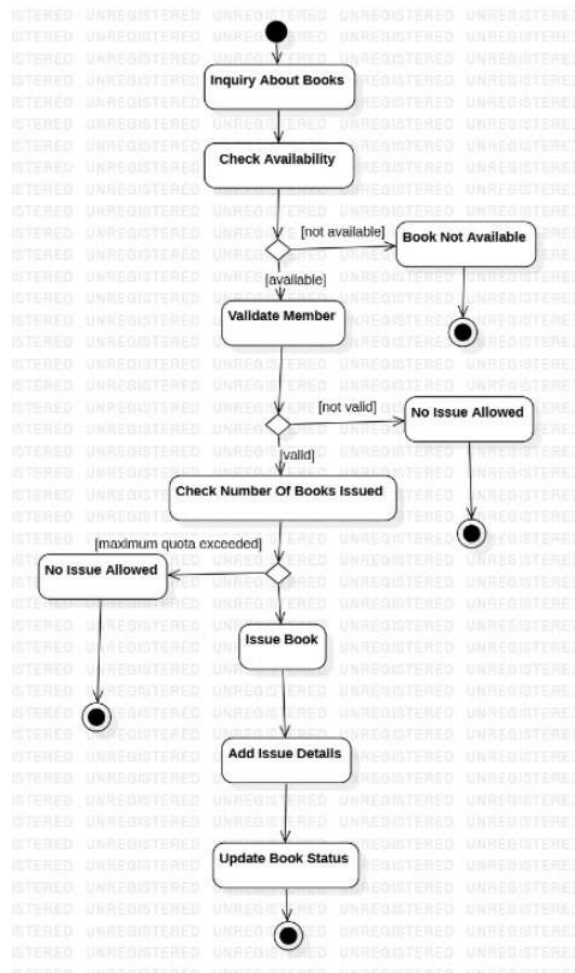
a. Use Case Model



b. Sequence Model

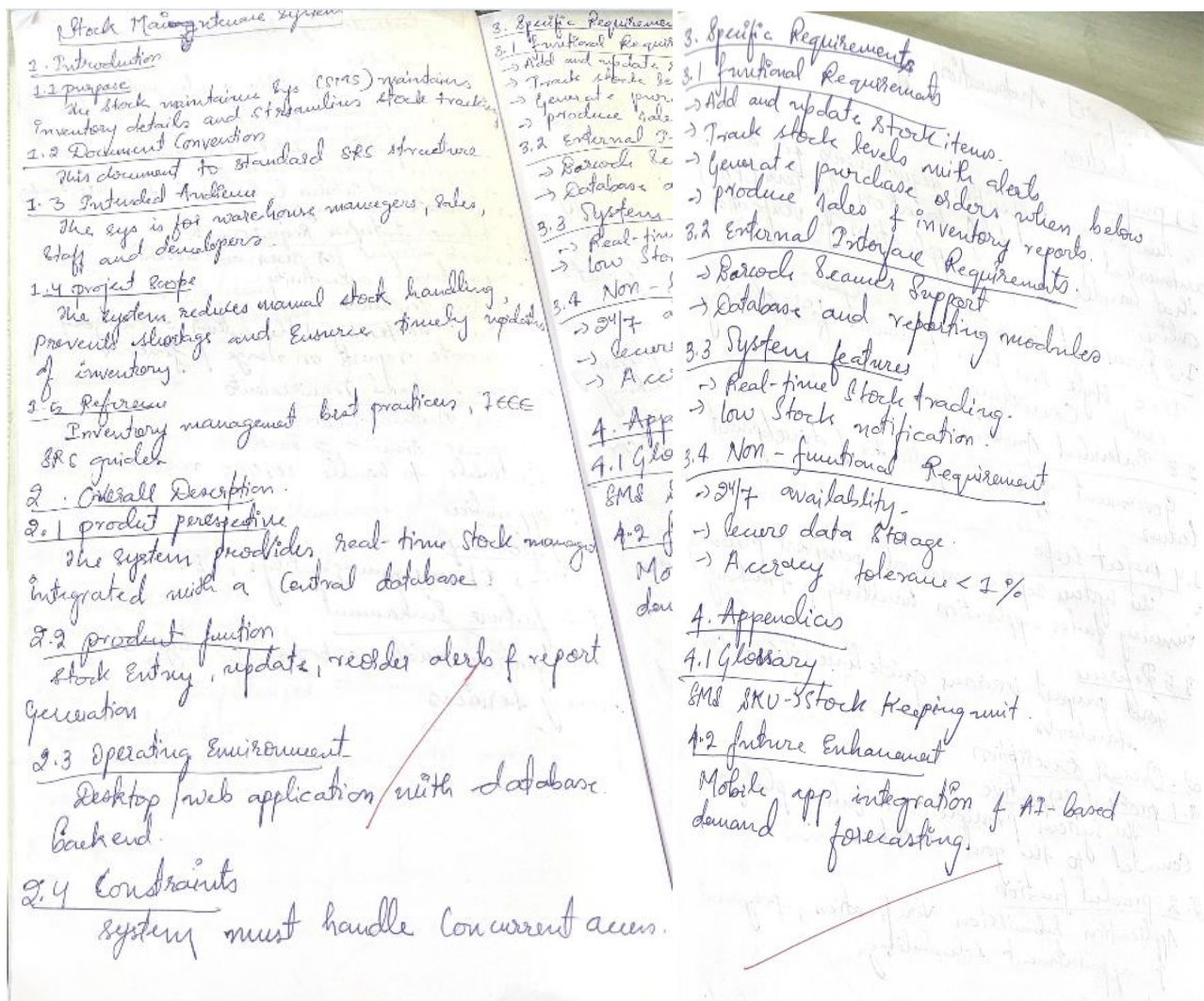


c. Activity Model

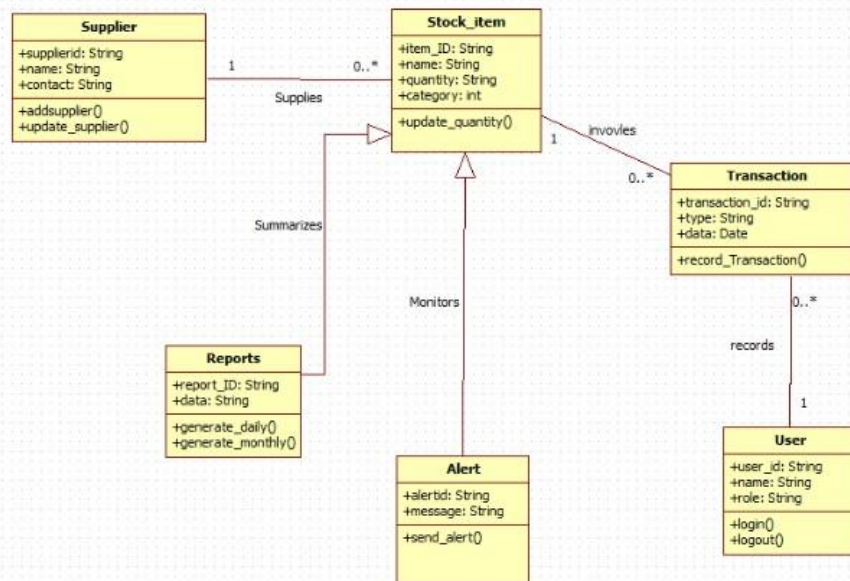


4. Stock Maintenance System

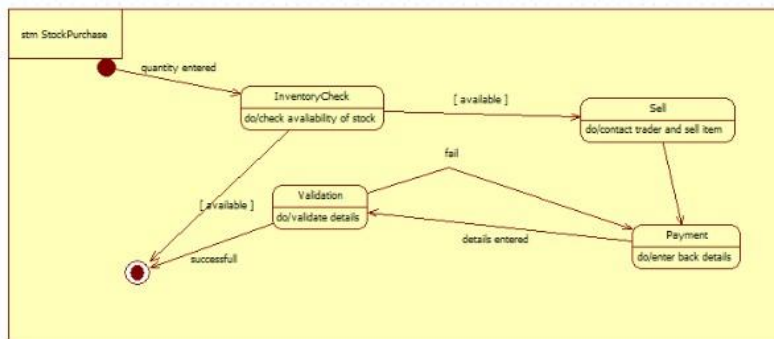
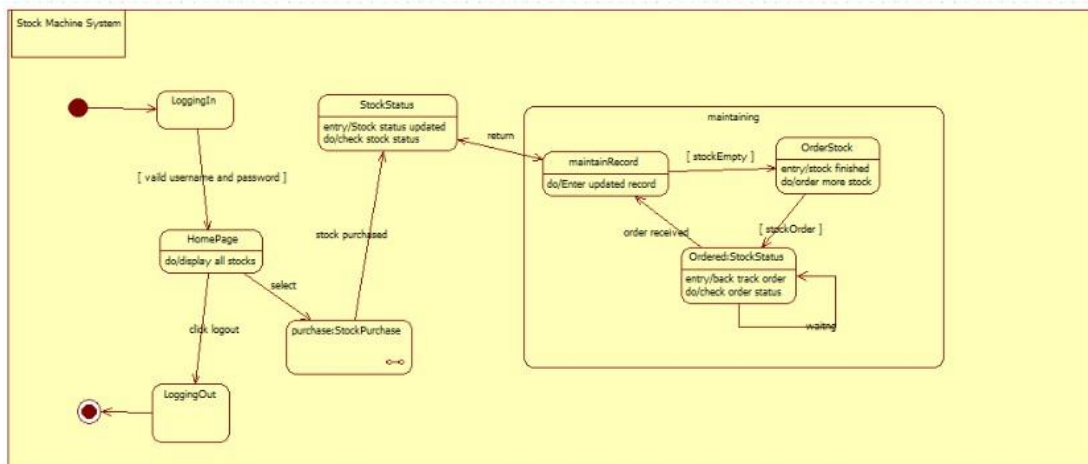
4.1 SRS Document



2.3 class model

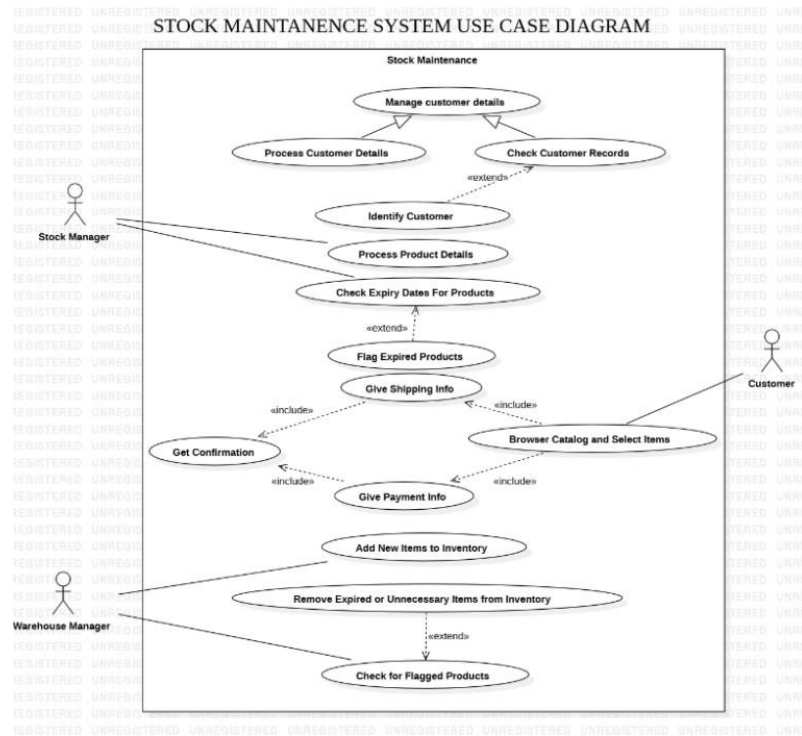


State Model

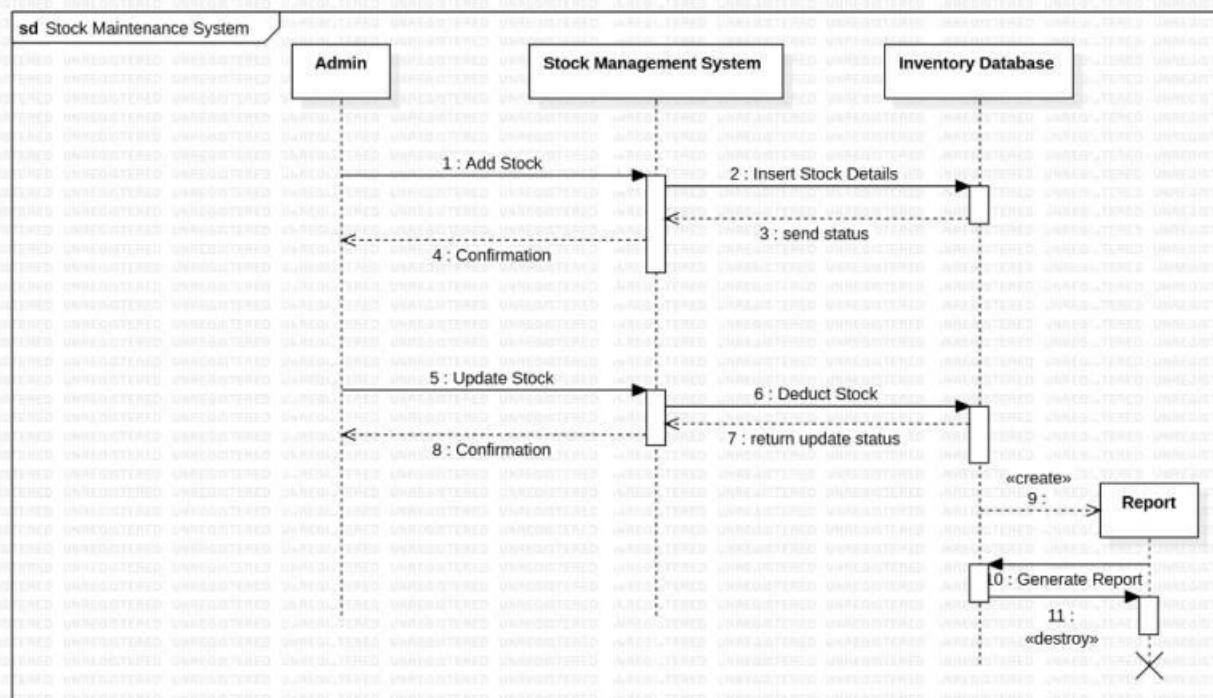


4.3 Interaction Models

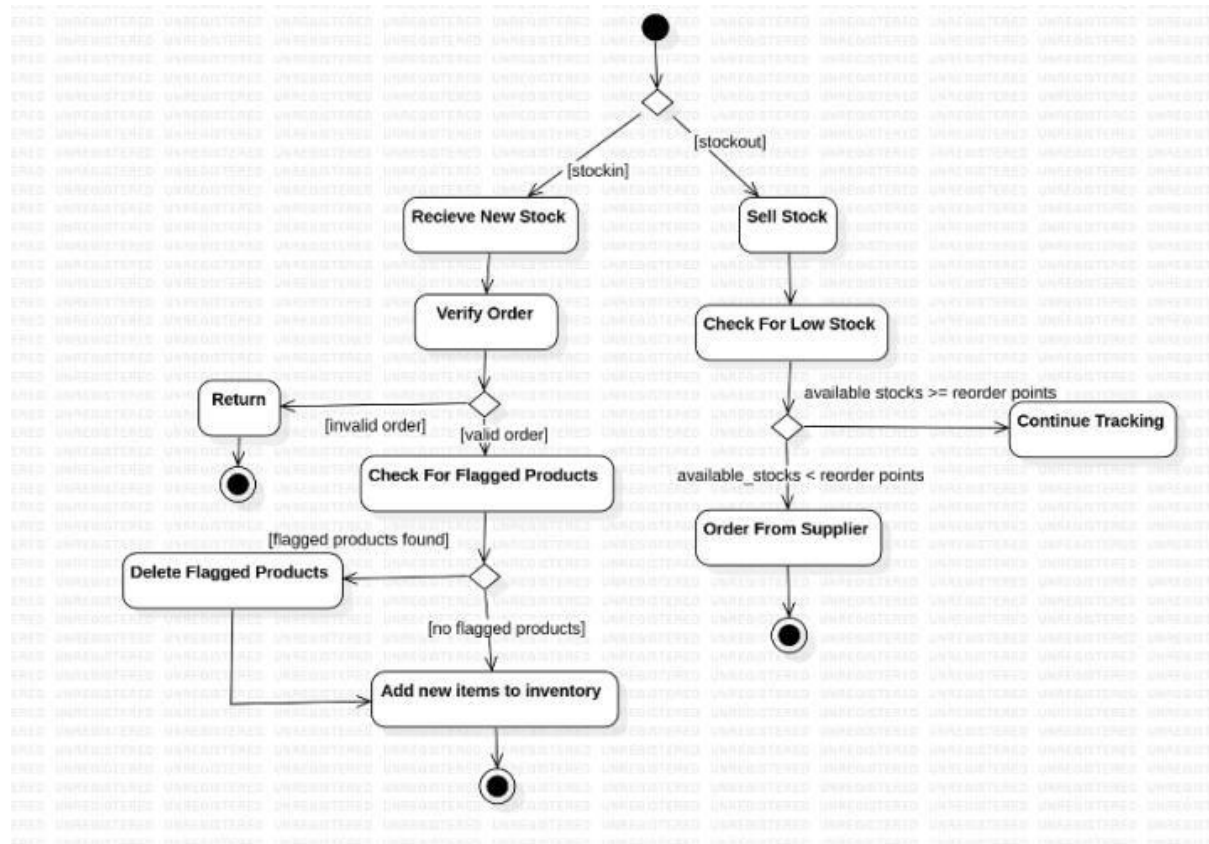
a. Use Case Model



d. Sequence Model



e. Activity Diagram



5. Passport Automation System

5.1 SRS Document

Passport Automation System

1. Introduction

1.1 Purpose
This document specifies requirements for an automated passport Automation System (PAS) that handles passport application & verification online.

1.2 Document Convention
IEEE style has been followed for clarity and consistency.

1.3 Intended Audience
Government officials, applicants & development teams.

1.4 Project Scope
The system replaces manual passport processing ensuring faster application handling & tracking.

1.5 Reference
Govt passport issuance guidelines IEEE SRS standards.

2. Overall Description

2.1 Product Perspective
The system provides an online platform connected to the govt database.

2.2 Product Function
Application submission, verification, payment & appointment scheduling.

2.3 User classes
Applicants, Verification officers & administration.

2.4 Operating environment
Web based system with secure authentication, connected to national database.

2.5 Constraints
Must comply with government regulations and data security policy.

2.6 Documentation
User guidelines and admin manuals will be supplied.

3.1 Automation Assumptions and Dependencies
Assume connectivity with police & national ID Database.

3. Specific Requirements

3.1 Functional Requirements

- Submit online application
- Upload required documents
- Schedule appointments
- Verify application details with police & govt records
- Process online payments.

3.2 External Interface requirement

- Web portal for users.
- Integration with payment gateways & government databases.

3.3 Non-functional Requirement

- High security for personal data
- 99.9% uptime
- Fast response within 3 sec

4. Appendices

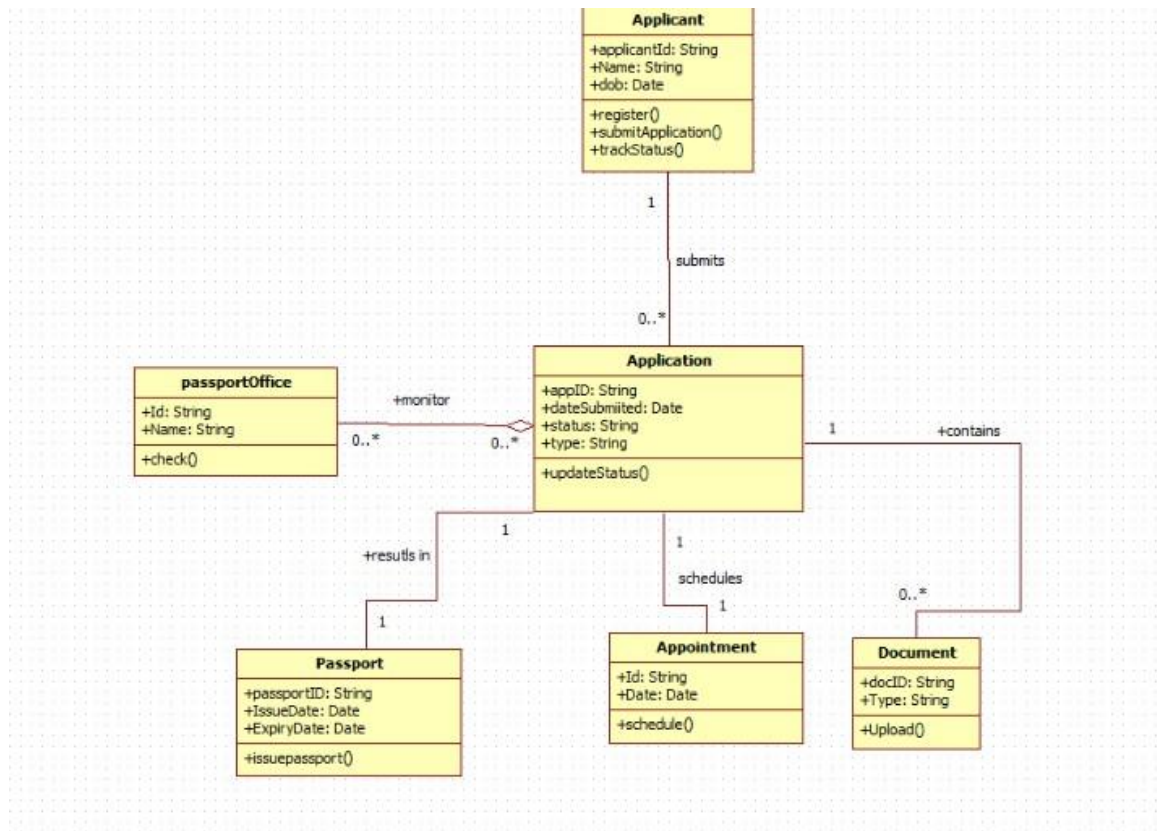
4.1 Glossary

- PAS → passport Automation System
- ID → Identification

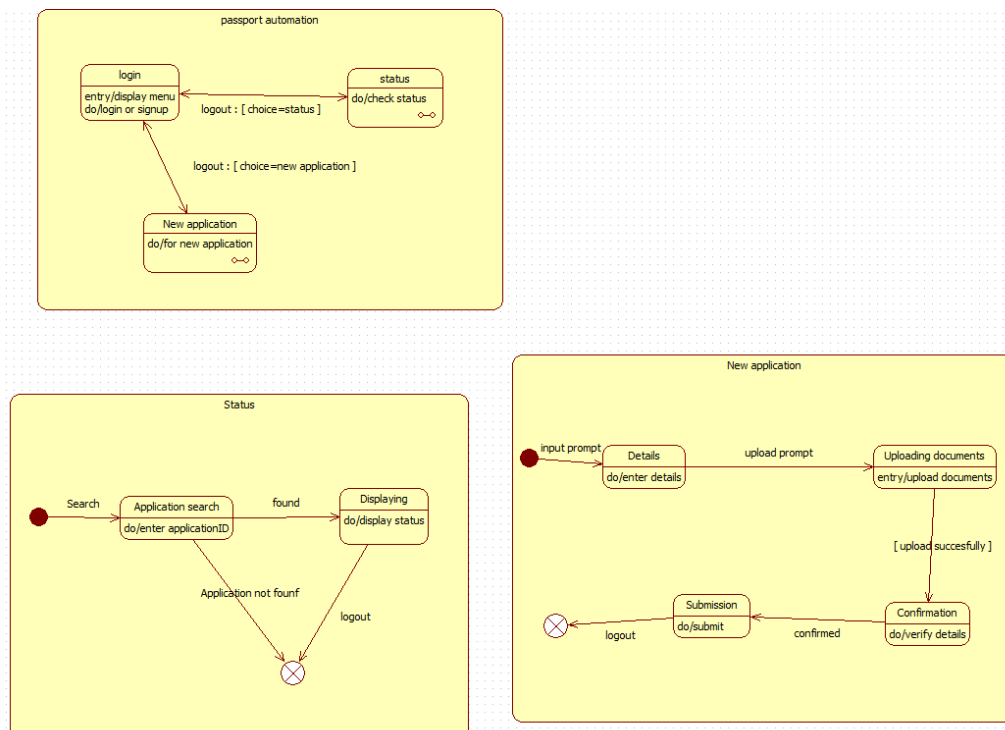
4.2 Future Enhancements

- Integration with Biometric Systems & mobile application.

Class Model

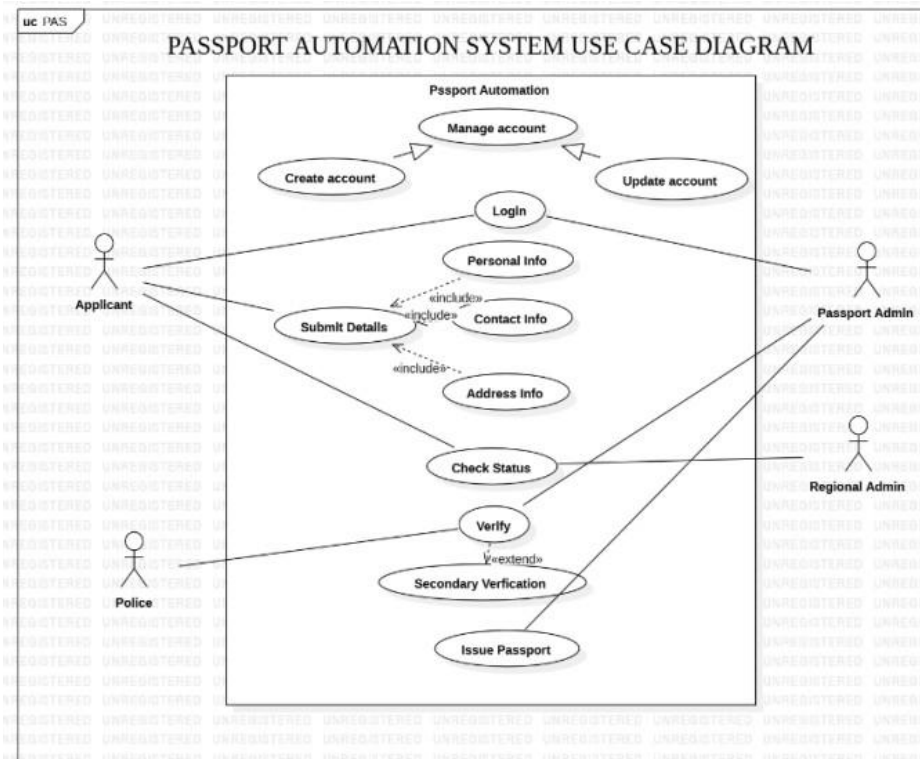


5.2 State Model

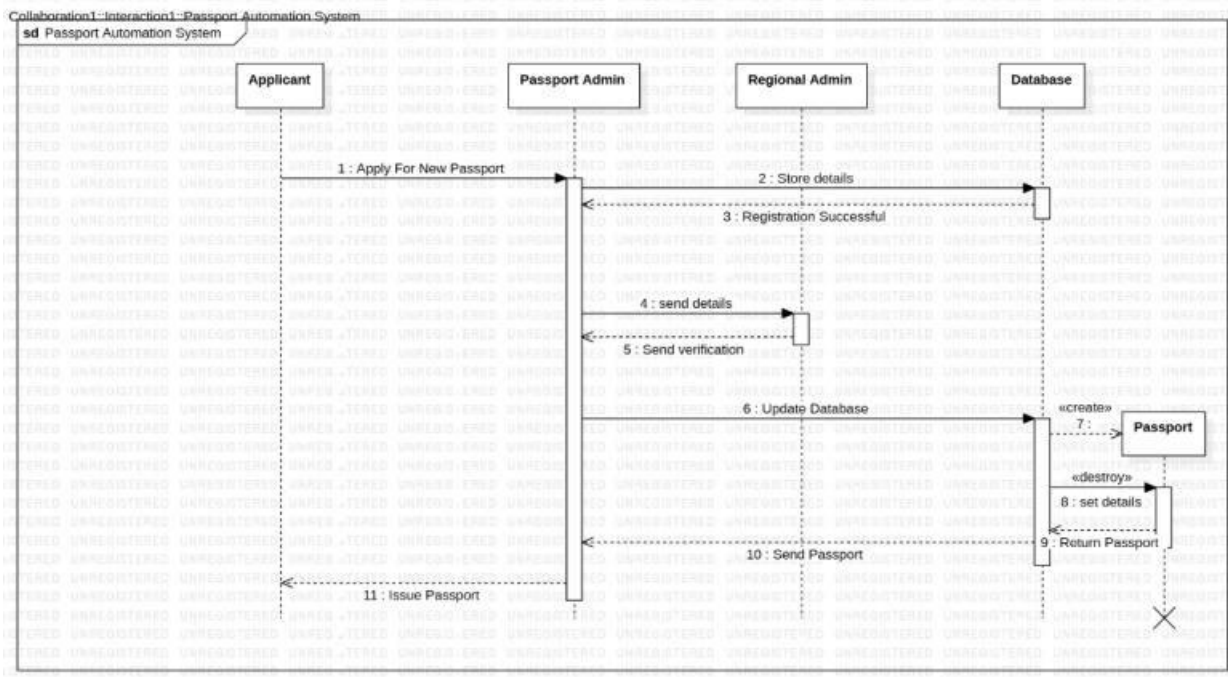


5.3 Interaction Models

a. Use Case Model



a. Sequence Model



b. Activity model

